Responses to the feedback on the thesis

I sincerely appreciate the referees' careful review and valuable comments.
I have tried to address all the suggestions and believe that the revised thesis has benefited substantially from the referee's feedback.

1. I have serious concerns regarding the overall quality, clarity, and flow of the thesis. The writing style and organization make the document difficult to read and, at times, frustrating to follow. Important details are not always easy to locate. For instance, Annexure 8.13—referred to multiple times across different chapters—could not be found in the submitted document. Similarly, Appendix 7.4.2 appears to be missing. Statements such as "The detailed process of designing the card deck is discussed in Section 3.3 of the publication are uncléar and confusing, particularly when the referenced material is not adequately presented within the thesis itself. There are several similar instances throughout the document.
   Additionally, the sequencing of content is often problematic. In some sections, interpretations are presented before the results are described, which disrupts the logical flow of the analysis. For example, on page 37 in Chapter 4, interpretations appear before Table 4.1, whereas the methods and results are discussed only after the table. Such inconsistencies in structure make it difficult to follow the argument and assess the findings systematically.

   Response:
   I acknowledge that certain sections might not have met the expected level of readability and coherence. I have revised the document, improved the writing style, and ensured smoother transitions between sections, enhancing overall organization and making the content easier to follow.
   Regarding Annexure 8.13 and Appendix 7.4.2, I have carefully reviewed the document to ensure that all referenced annexures and appendices are correctly included and consistently cross-referenced throughout the thesis. The table's positioning was due to LaTeX's default formatting, which disrupted the flow of information. It has now been revised and corrected.

2. My primary concern relates to the methodology of the study. The thesis does not clearly explain how the participants were selected. It is also unclear how they were allocated to the experimental and control groups. Was randomization used to assign participants to the intervention and control groups? If not, was any matching procedure adopted? Furthermore, was any attempt made to compare the baseline characteristics of participants across the two groups? Were the characteristics adequately balanced between the intervention and control groups? In the absence of such critical methodological details, it is difficult to assess the internal validity of the study. If the groups were not comparable at baseline, relying solely on a simple t-test would be insufficient to draw valid inferences. These methodological gaps significantly limit confidence in the reported findings. This concern is particularly relevant to Chapters 4, 5, and 6.

   Response:
   I acknowledge that the previous version does not clearly describe the recruitment and inclusion process. However, I have revised the methodology section on how participants were recruited, including any inclusion and exclusion criteria applied. Experimental and control groups were assigned based on availability and the sub-region they belonged to. Randomization was used in some locations (where there were enough participants), while in most cases, we used convenience sampling. I acknowledge that this introduces potential threats to internal validity. I have acknowledged this as a limitation. Regarding baseline comparability, I have included a table comparing demographic variables (such as years of schooling, last educational qualification, years of experience as a CHW, etc.) across groups to demonstrate baseline equivalence, which helped maintain a reasonable degree of comparability between groups. I have revised the Chapters to ensure that all methodological details are presented consistently and transparently, enabling a clearer assessment of the study's internal validity.

3. The thesis also employs qualitative methods; however, no clear justification is provided for their inclusion. It is not explained why Focus Group Discussions (FGDs) were considered necessary within the overall research design. The process of selecting participants for the FGDs is also unclear. Furthermore, the thesis does not describe how the FGDs were analyzed, whether any systematic analytical framework was adopted, or whether any qualitative data analysis software was used. Equally important, it is not evident how the insights generated from the FGDs were integrated into the broader analysis. The thesis does not clearly explain how the qualitative findings informed the interpretation of results or contributed to the overall research objectives. In the absence of these methodological details, it is difficult to interpret the qualitative findings with confidence.

   Response:
   Regarding the inclusion of Focus Group Discussions (FGDs), the purpose is to complement the quantitative findings by capturing participants' experiences, perceptions, and reflections that may not be fully captured by quantitative measures alone. Qualitative insights were intended to provide deeper contextual understanding and support the interpretation of observed outcomes.
   I acknowledge that this rationale was not sufficiently articulated in the previous version, but it has been revised now to clearly justify the inclusion of FGDs in Section 4.1 2a.
   There were no selection criteria; participation was voluntary, usually after the pretest or posttest.

4. In Chapter 4, the scholar states that "The AR-based game was more interesting and effective in retaining knowledge significantly, even after three months." However, it is unclear how knowledge retention after three months was assessed if the post-test was conducted immediately after the session. The thesis does not provide sufficient clarity regarding the timing and design of follow-up assessments to substantiate this claim.
   In the same chapter, it is also mentioned that "The AR-based game showed effective, immediate knowledge gains. However, the physical card game showed better knowledge retention." These statements appear to contradict the earlier assertion regarding the AR-based game's superiority in knowledge retention. The statistical analyses supporting these conclusions are not clearly presented, making it difficult to understand the basis for the interpretations discussed on page 38.

   Response:
   I acknowledge that the confusion was due to the way it was written. The intended interpretation is that the AR-based game resulted in stronger immediate learning gains, whereas the physical card game demonstrated comparatively better long-term retention. The statements are revised now.

5. Furthermore, in 'Study 1', the inclusion of a placebo or standard refresher training control group would have been essential to draw a robust conclusion that the card game was more effective than conventional refresher training methods. In the absence of such a comparison group, the validity of the claim that the card-based intervention is superior to standard training remains questionable.

   Response:
   I acknowledge that the corresponding t-test results are not mentioned in Table 5.1. This is a limitation in reporting rather than in the analysis itself. The CG, or control group, performed significantly worse than all other groups, nullifying the placebo effect ($p<0.05$). This confirms the RQ2. I acknowledge the absence of a clearly presented statistical analysis for this hypothesis. The main text, table, and charts are updated with results, p-values, and other details.

6. In Chapter 5, the scholar concludes that the difference between the digital card game and the physical card game groups was statistically significant. However, the corresponding t-test results supporting this claim are not

presented in Table 5.1. On the contrary, my reading of Table 5.1 suggests that there is no statistically meaningful difference between the digital and physical card game groups. This discrepancy raises concerns regarding the accuracy of the statistical analysis and the validity of the interpretations drawn from it. Furthermore, I was unable to locate any statistical analysis conducted to test Hypothesis H2. The absence of clearly reported results for a stated hypothesis is a serious omission and further undermines confidence in the analytical rigor of the thesis.

Response:
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7. The title of Section 6.5, "Pilot Testing Digital and Physical Card Game with Nursing Students," introduces nursing students into the thesis without adequate context or justification. There is no clear explanation as to why nursing students were included in the study or how their participation aligns with the overall research objectives, which primarily focus on CHWs. Moreover, despite the section title, much of the discussion centers on findings related to first- and second-year ANM trainees, with only limited reference to third-year B.Sc. nursing students. This inconsistency between the section title and its substantive content creates confusion. The rationale for including trainees—whether ANMs or nursing students—has not been clearly articulated, making it difficult to understand their role within the broader research framework.

   Response:
   The inclusion of nursing students was exploratory in nature. Nursing students share functional and educational overlaps with CHWs, particularly in frontline healthcare delivery, community engagement, and training needs. The aim was to check the applicability of the intervention across different levels of formal healthcare training. Their inclusion was intended to provide insights into how such game-based interventions could be integrated into early-stage training. I acknowledge that this rationale was not sufficiently articulated.

8. In addition to the major concerns outlined above, there are several minor issues that should be addressed before the thesis is considered for the award of the PhD degree. For instance, the name International Institute for Population Sciences should be spelled correctly in the thesis. Acronyms such as ASHAs and CHWs are used interchangeably without clarification, which may lead to confusion unless explicitly defined and distinguished at the outset.
   I also encountered statements that appear irrelevant to the core research. For example, the sentence "Aparajita Mondal's contributions to research are not a part of her academic pursuits" does not seem to add any value to the research presented in the thesis. Such statements should either be clarified in terms of their relevance or removed to maintain academic focus and coherence.

   Response:
   I acknowledge that there were some mistakes in spelling and acronyms. A spelling check has been conducted in both the main text and the references, and the use of acronyms is now consistent throughout the thesis.
   The committee suggested that compliance with co-authors' contributions be made explicit. Thus, a declaration has been made as clarification. To improve focus and clarity, the declaration has been moved from the ethics declaration (Chapter 3) to the Author's contribution section at the end.